

Palmer Penguins Mini Project

Anmol Sandhu

Introduction

This project examines morphological variation in penguins using the Palmer Penguins dataset. The primary objective is to assess whether the relationship between flipper length and body mass differs across species.

Reproducible research practices are essential for transparent and reliable scientific findings.

Research Question

Does the relationship between flipper length and body mass differ across penguin species?

Hypothesis

The relationship between flipper length and body mass differs among Adelie, Chinstrap, and Gentoo penguins. Specifically, species with longer flippers are expected to have greater body mass, but the strength of this relationship will vary among species.

Data and Methods

Data Source

Data were obtained from the `palmerpenguins` R package (Gorman et al., n.d.) and exported as a raw dataset.

Data Processing

The processed dataset was created by:

- Selecting species, flipper_length_mm, and body_mass_g
- Removing observations with missing values

The final processed dataset contains 342 observations and 3 variables.

Results

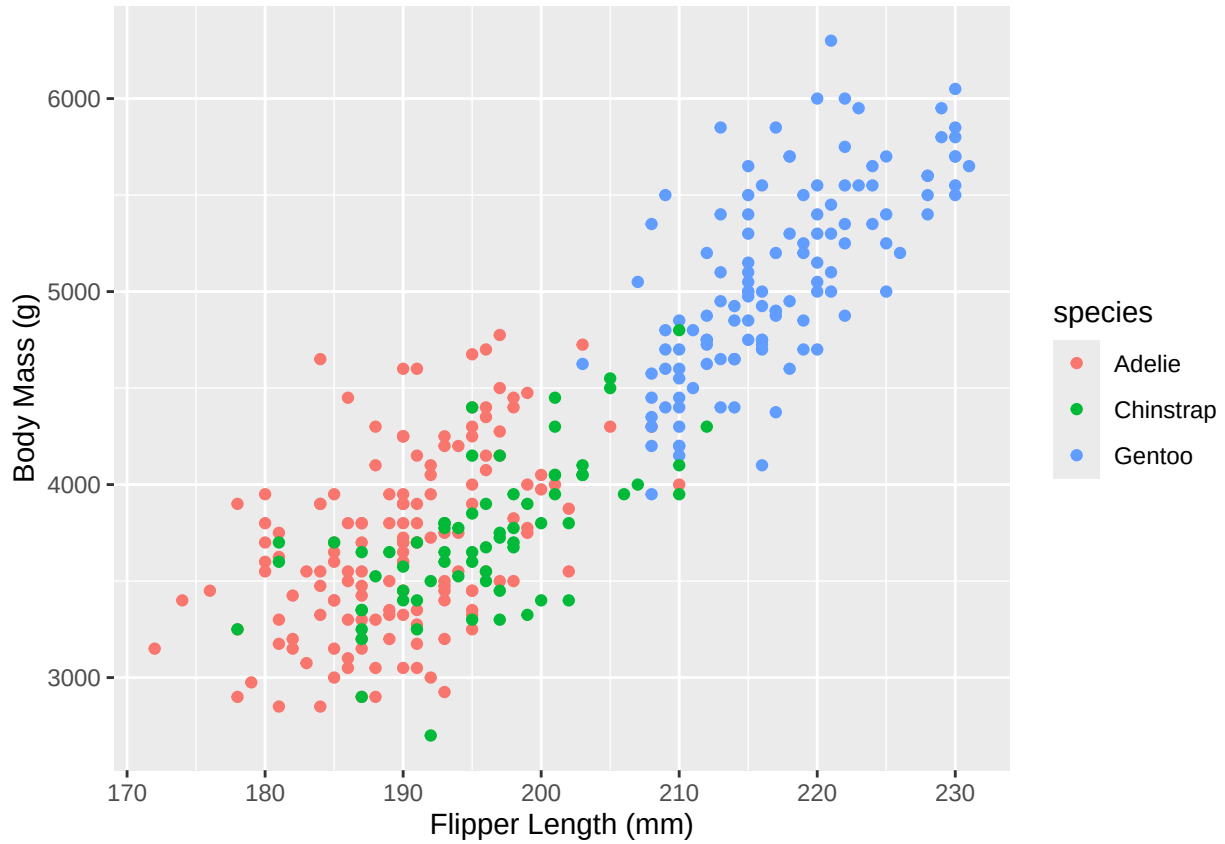


Figure 1. Relationship between flipper length and body mass across penguin species.

To evaluate whether the relationship between flipper length and body mass differs among species, a linear regression model including an interaction between flipper length and species was fitted.

Table 1: Linear regression model examining the interaction between flipper length and species on penguin body mass.

term	estimate	std.error	statistic	p.value
(Intercept)	-2535.836802	879.467667	-2.8833770	0.0041879
flipper_length_mm	32.831690	4.627184	7.0953936	0.0000000
speciesChinstrap	-501.358972	1523.459013	-0.3290925	0.7422908
speciesGentoo	-4251.443811	1427.332228	-2.9785944	0.0031064
flipper_length_mm:speciesChinstrap	1.741704	7.855734	0.2217112	0.8246734
flipper_length_mm:speciesGentoo	21.790812	6.941167	3.1393586	0.0018433

The interaction between flipper length and species was significant for Gentoo penguins ($p = 0.00184$), indicating that the relationship between flipper length and body mass differs among species.

These results indicate that the relationship between flipper length and body mass is not consistent across species. In particular, Gentoo penguins exhibit a stronger positive association compared to the other species.

Model assumptions including linearity, homoscedasticity, and normality of residuals were assessed and were found to be reasonably satisfied.

References

Gorman, K. B., Horst, A. M., & Hill, A. P. (n.d.). *Palmer archipelago (antarctica) penguin data*.